ARCHAEOPHD: TECHNICAL ARCHITECTURE OVERVIEW

Deterministic-Neural Dual Routing, Stratigraphic Graph Validation, and Air-Gapped Field Deployment

TARGET: **PhD Supervisor & Committee**	SCOPE: **Software Engineering Approach**	ARCHITECTURE: **Hybrid C++20 + Local LLM**

CORE ENGINEERING THESIS:
Archaeological monograph ingestion demands 100% numerical fidelity, mathematical graph cycle detection, and strict data confidentiality under state antiquities permits. Generic cloud LLMs fail on silent numerical drift and regulatory secrecy; pure regex systems fail on irregular 19th-century prose syntax. ArchaeoPhD resolves this through a deterministic C++ fast path backed by a strictly governed, air-gapped local neural arbitrator.

1. REJECTION OF GENERIC CLOUD AI & PURE REGEX SYSTEMS

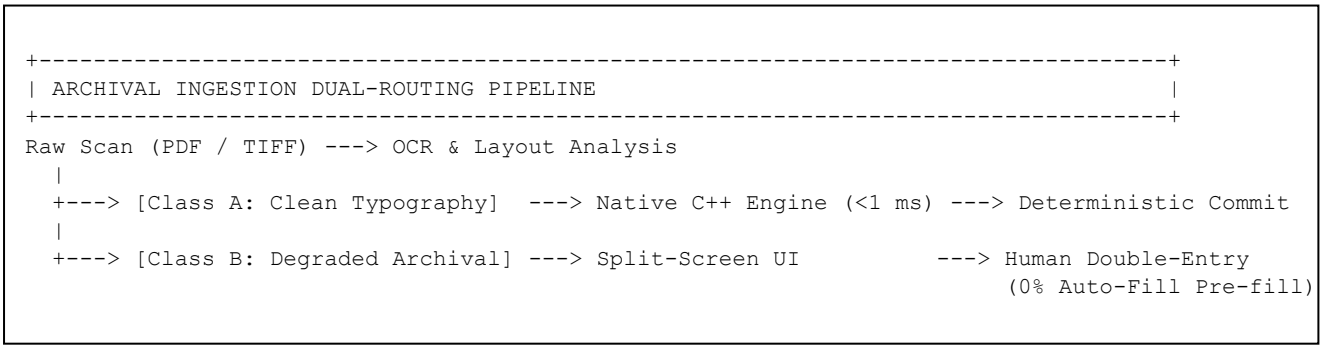
FAILURE MODES OF GENERIC CLOUD AI (GPT-4 / CLAUDE)

- **Probabilistic Hallucination & Numerical Drift:** In archival OCR benchmarks, commercial LLMs silently alter critical excavation coordinates and depth strata (empirical test transcribed *1966* as *1656*, and *20–40 cm* as *20–10 cm*). Numeric fidelity is non-negotiable.
- **Antiquities Permit NDAs & Looting Risk:** Excavation field reports contain unexcavated findspot GPS coordinates. Sending unredacted data to cloud APIs violates state antiquities permits and exposes vulnerable sites to illicit looting.
- **Prohibitive Latency:** Full-page cloud LLM inference takes 15–30s per page, making batch ingestion of 500-page monographs computationally unviable.

FAILURE MODES OF PURE REGEX / TRADITIONAL PARSERS

- **Multi-Clausal Syntactic Collapse:** Regular expressions fail catastrophically when a single sentence contains multiple chronological layers (e.g., *"pottery recovered in 1968, though Evans (1921) noted similar forms dating to 1450 BCE"*).
- **Multilingual Archaic Variations:** Fails across historical German, French, and Victorian English excavation prose where syntactic structures and abbreviations diverge unpredictably.

2. THE CHOSEN ARCHITECTURE: HYBRID C++20 + GOVERNED LOCAL NEURAL MODEL



- **Native C++20 Fast Path:** Executes deterministically in sub-millisecond time with zero heap allocation per token, extracting measurements, counts, and stratigraphic relationships via mathematical grammar parsers.
- **Governed Local Neural Model:** An in-process, 4-bit quantized Qwen 2.5 7B GGUF (~4.4 GB) model, invoked solely for semantic ambiguity resolution. It runs fully offline with zero internet access required.
- **Strict Air-Gap Guarantee:** The system opens zero outbound TCP/UDP ports. All UI communication executes through in-memory Win32 WebView2 IPC, ensuring complete compliance with antiquities permit NDAs in remote field excavation camps.

1. ALGORITHMIC DIVISION OF LABOR MATRIX

TASK / DOMAIN SUBSYSTEM	ENGINE DEPLOYED	ALGORITHMIC IMPLEMENTATION & RATIONALE	BENCHMARK
Physical Measurements Depths, elevations, dimensions	`C++20 Engine`	Deterministic grammar enforcing regular units (m, cm, mm, BP, %). Enforces 0% numeric loss invariant.	`< 1.0 ms / p`
Artifact Count Binding Sherds, bifaces, lithics	`C++20 Engine`	Rejects bare numbers unless syntactically bound to verified head nouns (e.g., *"14 rim sherds"*).	`< 0.2 ms`
Author Lifespan Masking Bibliographic suppression	`C++20 Engine`	Strips biographical date ranges (e.g., *Petrie (1853–1942)*) to prevent citation dates displacing strata.	`< 0.1 ms`
Stratigraphic Harris DAG Steno's Law physical sequencing	`C++20 Engine`	Tarjan's Strongly Connected Components (SCC) O(V+E) detects impossible physical cycles; Kahn's sort.	`< 5.0 ms (1k)`
Radiocarbon Splines 14C non-linear calibration	`C++20 Engine`	Continuous piecewise cubic splines on IntCal20 calibration curves (1σ / 2σ probability envelopes).	`< 0.1 ms`
Hybrid Retrieval Dense vector + exact lexical	`C++20 Engine`	128-dim Matryoshka dense vectors fused with in-process Okapi BM25 (k=60) to prevent keyword drift.	`22.5 ms`
Clausal Disambiguation Multi-date sentence context	`Qwen 2.5 7B`	Disambiguates competing historical vs. event dates in complex multi-clausal sentences in excavation prose.	`~3.8 s / call`
Historiographical Conflicts Opposing excavation hypotheses	`Qwen 2.5 7B`	Identifies methodological disputes and chronology conflicts across publications via local KV-cached inference.	`~4.2 s (cache)`

2. GOVERNED NEURAL ARBITRATION: HOW HALLUCINATIONS ARE ELIMINATED

THE SELECTION-ONLY GUARDRAIL ARCHITECTURE

Unlike consumer chat interfaces, ArchaeoPhD never permits the neural model to generate free-form text:

- **Grammar-Constrained Output:** The model receives the verbatim text snippet alongside the pre-extracted C++ candidate list. It is constrained via grammar sampling to output *only* the index integer of the verified candidate or `REJECT`. It cannot invent names or dates.
- **The Class B Invariant:** *"A blank field is infinitely safer than a plausible wrong one."* On degraded 19th-century letterpress scans, automated pre-fill is locked out; the researcher transcribes via split-screen verification.

1. CRITICAL FAILURE MODES & ENGINEERING MITIGATIONS

- **Where C++ Deterministic Rules Fail:** Syntactic parsers cannot resolve unaligned multi-column OCR tables or damaged physical plates where typography is fragmented.
  Mitigation: Flagged deterministically as `UNANCHORED_OR_DEGRADED` and routed to the human verification queue.
- **Where the Local Neural Model Fails:** 3–5s execution latency and 4.4 GB RAM footprint. If unconstrained, it risks plausible confabulations.
  Mitigation: Strict quarantine to multi-candidate arbitration; constrained index decoding prevents free-form generation.
- **Where Dense Semantic Vector Retrieval Fails:** High-dimensional embeddings cluster documents using similar theoretical terminology even if their stratigraphy or geographic regions conflict.
  Mitigation: Dual-stage lexical BM25 filtering enforces exact trench identifiers, locus numbers, and site names.

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Hardware Footprint Verification:
• Standalone Win32 Executable: 9.49 MB (release/ArchaeoPhD.exe)
• Neural Model Weights:        4.45 GB total (Qwen2.5-7B-Instruct-Q4_K_M.gguf + nomic-embed-text)
• Runtime Footprint:           ~65 MB base idle RAM | Opens 0 network sockets | Fully Air-Gapped
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2. PLANNED TECHNICAL EXPERIMENTS (DOCTORAL RESEARCH ROADMAP)

EXPERIMENT 1: LOCAL VISION-LANGUAGE MODELS (VLMS) ON HIGH-RESOLUTION LETTERPRESS CROPS

Engineering Goal: Test quantized local VLMs (Qwen2-VL 2B/7B GGUF) directly on segmented typographic plates from 19th-century monographs (e.g., Petrie, Evans). By processing high-resolution bounding crops rather than full pages, we eliminate token collapse and preserve sub-millimeter plate coordinates without cloud data leakage.

EXPERIMENT 2: 3D VOLUMETRIC VOXEL INTERSECTIONS IN THE HARRIS MATRIX DAG

Engineering Goal: Bridge trench Structure-from-Motion (SfM) 3D photogrammetric meshes directly into the C++ Tarjan DAG solver. Instead of relying on manual 2D context sheets, the engine will compute 3D volumetric intersections between stratigraphic units to mathematically detect physical overlap violations.

EXPERIMENT 3: NATIVE C++ BAYESIAN SEQUENTIAL CALIBRATION ENGINE (OXCAL-IN-C++)

Engineering Goal: Embed an in-process Markov Chain Monte Carlo (MCMC) Bayesian sampler directly into the native C++ binary. This fuses prior stratigraphic DAG topological constraints with posterior IntCal20 continuous radiocarbon probability envelopes on-device, replacing external commercial legacy software.

SUMMARY FOR THE DOCTORAL ADVISORY COMMITTEE:

ArchaeoPhD pairs deterministic C++ engines for sub-millisecond mathematical precision and zero numerical drift with a governed, offline neural model for semantic nuance. The resulting architecture delivers a fully air-gapped, verifiable computational platform engineered for doctoral-level research.